

Classwork 04//LSTM

Student Name: _____ **Points:** 30

A Number: _____ **Due Date:** Feb 19, 2026

Assume

- W_f, b_f weights and bias for the forget gate, U_f weights for recurrent connection for the forget gate.
- W_i, b_i weights and bias for the input gate, U_i weights for recurrent connection for the input gate.
- W_o, b_o weights and bias for the output gate, U_o weights for recurrent connection for the output gate.
- c_t is the cell state or the context vector at time t (like a memory tape).
- W_c, b_c weights and bias for the cell state, U_c weights for recurrent connection for the cell state.

Forget gate operation:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

Input gate operation:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

Updated cell state:

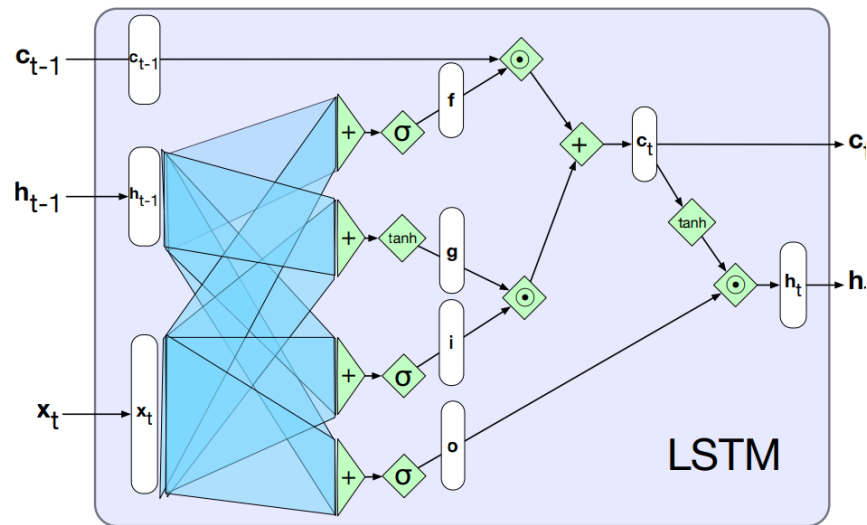
$$\begin{aligned} \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \cdot \tilde{c}_t \end{aligned}$$

Output gate:

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

Hidden state:

$$h_t = o_t \odot \tanh(c_t)$$



1 LSTM Output Calculation

Initial input $x_t = [0.5]$

Previous hidden state $h_{t-1} = [0.2]$

Previous cell state $c_{t-1} = [0.8]$

1. For the forget gate, consider $W_f = [0.5]$, $U_f = [0.3]$, $b_f = 0.0$ Compute the forget gate output f_t .
2. For the input gate, consider $W_i = [0.6]$, $U_i = [0.4]$, $b_i = 0.0$ Compute the input gate output i_t .
3. For the candidate cell, consider $W_c = [0.4]$, $U_c = [0.5]$, $b_c = 0.0$ Compute the updated cell state c_t .
What percentage of old memory is kept?
What percentage of new candidate is kept?
4. For the output gate, consider $W_o = [0.7]$, $U_o = [0.6]$, $b_o = 0.0$ Compute the input gate output o_t .
5. Compute the final hidden state h_t .